Ecology

Subject: Biology | NoteShare Library

ECOLOGY

1. LEVELS
Organism -> Population -> Community
-> Ecosystem -> Biome -> Biosphere

2. ECOSYSTEM
Biotic: Producers, Consumers, Decomposers
Abiotic: Sun, Water, Temp, Soil
Energy flow: 10% rule
Food chain, Food web
Trophic levels 1-4

3. CYCLES
Water: Evaporation, Condensation, Precipitation
Carbon: Photosynthesis, Respiration, Combustion
Nitrogen: Fixation, Nitrification, Denitrification
Phosphorus: Rock weathering

4. POPULATION
Exponential: dN/dt = rN
Logistic: dN/dt = rN(K-N)/K
Survivorship: I, II, III
Age structure

5. COMMUNITY
Competition (-/-)
Predation (+/-)
Mutualism (+/+)
Commensalism (+/0)
Parasitism (+/-)
Succession: Primary, Secondary

6. BIOMES
Tropical Rainforest
Savanna, Desert
Temperate Grassland, Deciduous
Taiga, Tundra

7. ENVIRONMENT
Pollution, Climate Change
Ozone, Deforestation

Conservation